

Highlights

Terrain-Aware Input Representations for Learning-Based Real-Time Wind Field Prediction

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- Flow incidence angle feature explicitly encodes terrain-wind directional interactions
- Polar wind encoding gives continuous angular representation, no discontinuities
- Architecture-agnostic; tested on 3DNN, FNO, U-FNO, U-Net at no inference cost
- Flow incidence angle matches or beats baseline on nearly every metric and dataset
- Polar encoding gives largest CFD gains but is sensitive to terrain-scale shift

Terrain-Aware Input Representations for Learning-Based Real-Time Wind Field Prediction

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Abstract

Real-time, high-resolution wind field estimation is critical for autonomous UAV operation in complex near-ground environments. Recent work has demonstrated that deep neural networks trained on Computational Fluid Dynamics (CFD) data can predict wind fields from sparse measurements and terrain geometry, enabling fast inference suitable for onboard deployment. However, most existing wind prediction methods encode terrain and wind independently, requiring networks to implicitly learn terrain-flow interactions from data. In this paper, we propose two complementary input features: a flow incidence angle feature encoding the angular relationship between the incoming wind direction and the local terrain gradient, and a polar representation providing a continuous angular encoding of wind velocity. These features are architecture-agnostic and introduce an inductive bias that improves prediction accuracy while incurring negligible computational overhead. We evaluate the proposed features across four neural architectures spanning convolutional and spectral paradigms (3DNN, FNO, U-FNO, U-Net) on CFD simulations, UAV flight trajectories, and three near-surface mast campaigns. The flow incidence angle improves or matches the baseline on essentially every metric across all four architectures and all three evaluation settings, while the polar encoding delivers larger gains on CFD and on the in-distribution Perdigão mast campaign but is sensitive to terrain-scale shift on the smaller Bolund and Askervein campaigns.

Keywords: wind field prediction, UAV, deep learning, terrain-aware

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1. Introduction

Safe and efficient operation of UAVs in obstacle-rich outdoor environments containing urban structures and complex terrain is crucial for many real-world applications, including autonomous delivery, search and rescue missions, and military operations. The ability to operate safely in such environments depends critically on accurate knowledge of local wind fields, which can vary sharply due to terrain-induced flow phenomena. Wind-aware planning and control require volumetric wind information: energy-optimal path planners exploit non-uniform wind fields to extend range or endurance, wind-preview model-predictive controllers use lookahead wind to pre-compensate for gusts and shear, and safety-aware planners avoid lee-side recirculation and ridge-crossing turbulence. Folk et al. [1] demonstrate the closed-loop value of local wind information for urban UAV navigation, reporting up to 30% lower energy consumption and a 32% reduction in failure rate when a receding-horizon planner is given access to the local time-averaged wind field.

Recent learning-based methods [2, 3] have achieved promising results in real-time wind field prediction from sparse measurements and terrain information. However, these approaches encode wind information in a global Cartesian coordinate frame independent of terrain geometry, leaving the directional relationship between flow and terrain to be learned implicitly. Additionally, the need for real-time inference on UAV hardware constrains model size, potentially limiting expressiveness.

We focus on improving wind field prediction accuracy from sparse onboard UAV time-averaged wind measurements without increasing computational complexity or relying on online CFD solvers. Our goal is to design input representations that more effectively leverage known terrain structure and onboard measurements to capture terrain-induced wind variations in real time.

We propose two simple, physically motivated modifications to the input representation: a flow incidence angle feature encoding terrain-flow interactions, and a polar parameterization of wind measurements. These changes are architecture-agnostic, incur negligible computational overhead at inference, and introduce an inductive bias toward physically consistent predictions. As

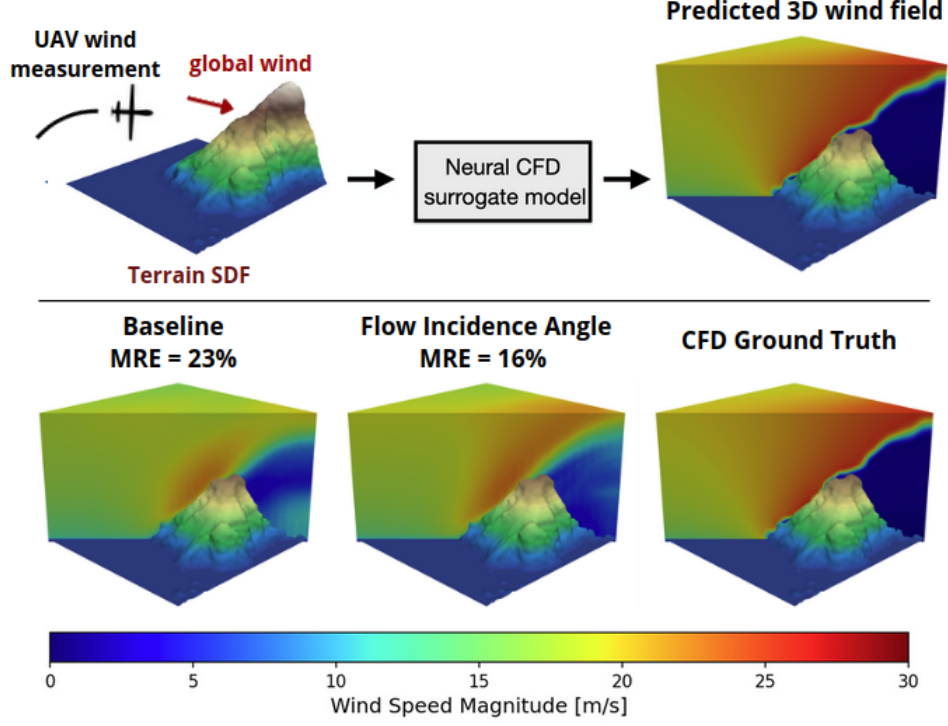


Figure 1: Overview of neural CFD-based wind field prediction. Top: a surrogate model takes terrain geometry (represented as a signed distance field) and sparse UAV wind measurements as input, and predicts the full 3D time-averaged wind field together with the turbulent-kinetic-energy field. Bottom: qualitative comparison between the baseline model [2] and our proposed variant incorporating the flow incidence angle (α) feature. Both models receive identical inputs; the sole modification is the inclusion of the α feature encoding terrain-flow interactions. The proposed method better captures the corner-acceleration region around the windward edge of the mountain and the recirculation region, reducing the relative mean absolute error from 23% to 16% of the domain-mean wind speed at negligible additional inference cost.

illustrated in Figure. 1, the proposed features improve prediction accuracy, particularly in terrain-affected regions.

The main contributions of this paper are:

- (i) two complementary input features: a flow incidence angle feature encoding windward and leeward relationships via the angle between the terrain gradient and the wind direction, and a polar representation of the wind velocity that avoids discontinuities in angular encoding,

- (ii) a systematic evaluation across four architectures spanning convolutional and spectral paradigms (3DNN, FNO, U-FNO, U-Net), demonstrating that the proposed features are architecture-agnostic,
- (iii) evaluation on CFD simulations, UAV flight trajectories, and three near-surface meteorological mast campaigns spanning an order of magnitude in terrain scale, identifying the flow incidence angle as the safest single feature and characterising the terrain-scale sensitivity of the polar encoding.

2. Related Work

Deep learning surrogates for wind- and fluid-field prediction span a wide range of architectures. Convolutional U-Nets remain a strong baseline for gridded inputs [2, 4, 5], exploiting multi-scale skip connections to capture both global flow context and local terrain-induced features. Fourier neural operators (FNO) [6] learn mappings between function spaces via spectral convolutions, providing discretization-invariant predictions but exhibiting blurring on sharp local gradients. U-FNO [7] and Local-FNO [8] address this limitation by combining FNO’s spectral efficiency with multi-scale or local-patch decompositions that better resolve flow-feature edges and small-scale turbulence. Geometry-aware operators extend these ideas to irregular, non-grid domains: GINO [9] processes graph-based representations of arbitrary geometries via signed-distance fields and graph-kernel encoders that bridge to regular FNO latents, while Transolver [10] uses physics-attention over learned domain partitions to handle general meshes at linear cost. At the atmospheric scale, vision-transformer architectures such as Pangu-Weather [11] and FourCastNet [12] have shown that attention-based operators scale effectively to volumetric atmospheric prediction. Within the wind-prediction literature specifically, terrain-aware downscaling networks reconstruct fine-resolution wind from coarse numerical-weather-prediction outputs: Wind-Topo [13] and DEVINE [14] are the foundational references in this lineage. DEVINE is particularly relevant to the present paper: it rotates each DEM patch around its central pixel relative to the inflow direction before passing it to the network, effectively a discrete approximation of the flow-incidence-angle encoding we propose, applied at the input geometry rather than as an additional channel. Most recently, Lin et al. [15] introduced FuXi-CFD, a Vision Transformer that downscales kilometer-scale wind to 30 m 3D fields.

FuXi-CFD targets server-side downscaling rather than onboard inference. An alternative paradigm targets onboard UAV reconstruction from sparse sensors: WindSeer [2] uses a U-Net on sparse wind measurements with a terrain SDF, while Folk et al. [3] predict local urban wind fields with an MLP from LiDAR range sensing and a single co-located wind measurement. Complementary to these data-driven approaches, physics-informed methods enforce governing equations as soft constraints: Yan et al. [16] and Chen et al. [17] embed RANS or LES residuals into PINNs for wind-field assimilation from sparse LiDAR.

These methods differ in their source of wind information: mesoscale-to-microscale downscalers [13, 14, 15, 5, 18] use dense numerical-weather-prediction (NWP) outputs as input, while sparse-reconstruction approaches [2, 3] infer the dense field from a small number of in-situ measurements. Across this diversity, input representations have remained largely consistent. Terrain geometry is typically encoded as raw elevation maps [4, 5, 13, 14, 15] or signed distance fields [2, 19]. Wind information is provided as Cartesian velocity components, either as sparse point measurements [2, 3] or gridded fields [15, 18, 5]. Graph-based approaches [20, 21] encode spatial relationships between sensors and their local environment through engineered graph features that combine geometric distances and terrain or obstacle descriptors. These works inject geometric structure through the graph topology while leaving the wind representation in standard Cartesian form; our contribution is orthogonal, modifying the wind and terrain representations themselves as scalar channels usable by any architecture. To our knowledge, no prior work has investigated input features that explicitly encode the directional relationship between wind and terrain geometry.

Sparse wind measurements at the vehicle can be obtained either directly, via dedicated flow sensors (pitot-static tubes [22], airflow vanes [2], and multi-hole pressure probes [23]) or indirectly, by inferring wind from the aircraft’s dynamic response using standard autopilot sensors and model-based estimators [24, 25, 26]. Direct sensing offers higher accuracy at the cost of dedicated hardware; indirect estimation reuses existing sensors but depends on accurate aerodynamic models. A comprehensive treatment is given by Tomić et al. [27]. Our focus is not on obtaining these measurements but on reconstructing dense wind fields from them.

An alternative paradigm bypasses explicit wind estimation entirely and rejects wind disturbances at the controller level. Adaptive geometric controllers [28] and incremental nonlinear dynamic inversion (INDI) [29] use online adaptation or high-rate inertial feedback to absorb gust loads without an explicit wind model. These methods stabilise flight under unknown wind but produce only information local to the vehicle, whereas wind-aware planning needs the wind field over the lookahead domain, which is the regime targeted by this work.

3. Theory

Our approach assumes prior knowledge of the terrain geometry and sparse, time-averaged wind measurements within the domain of interest. The objective is to reconstruct the dense, time-averaged wind field $\hat{\mathbf{U}}$ and the corresponding turbulence kinetic energy field $\hat{\mathbf{K}}$ around the terrain using sparse wind observations $\mathbf{W}_{\text{meas}}$ and terrain information $\mathbf{T}$. The proposed formulation closely follows the approach introduced in the WindSeer framework [2].

The problem is cast as a three-dimensional flow field reconstruction task, which consists of learning a mapping

$$(\hat{\mathbf{U}}, \hat{\mathbf{K}}) = f_{\theta}(\mathbf{T}, \mathbf{W}_{\text{meas}}), \quad (1)$$

where $\mathbf{W}_{\text{meas}}$ denotes sparse, time-averaged wind measurements within the domain, $\mathbf{T}$ represents the encoded terrain geometry, and f_{θ} is a learnable neural network parameterized by θ .

3.1. Input Representations

The input to the network is a three-dimensional uniform tensor

$$\mathbf{X} \in \mathbb{R}^{4 \times H \times W \times D}. \quad (2)$$

Following the baseline formulation of [2], the input comprises sparse horizontal wind measurements (u, v) expressed in a global Cartesian frame, a binary mask indicating observation locations, and terrain geometry encoded as a signed distance field (SDF). In this work, we propose two extensions to this representation.

Flow Incidence Angle Encoding. To explicitly encode the directional relationship between the wind field and the surrounding terrain, we introduce the *flow incidence angle* α , defined as the angle between the local terrain gradient $\nabla\mathbf{T}$ and the wind direction. Although this information is implicitly contained in the Cartesian wind components and the SDF, encoding it explicitly provides a direct indication of windward ($\alpha < 90^\circ$) and leeward ($\alpha > 90^\circ$) regions, corresponding to areas in front of and behind terrain features, as can be observed in Figure 2. The physical significance of this angle is well-established in CFD: flow incidence determines not only where boundary layer separation occurs on curved surfaces [30, 31], but also the magnitude of terrain-induced speed-up, which follows an approximately cosine relationship with approach angle [32]. Field measurements at Bolund Hill show that wake height varies by 10–70% when wind direction shifts by just $\pm 15^\circ$ [33], while large-eddy simulations confirm that identical ridges produce fundamentally different flow structures under different approach directions [34]. These findings motivate encoding flow incidence angle as an inductive prior that provides the network with direct information about the flow-terrain interaction regime.

The flow incidence angle α is defined as

$$\alpha = \arccos\left(\frac{\nabla\mathbf{T} \cdot \mathbf{u}}{\|\nabla\mathbf{T}\| \|\mathbf{u}\|}\right), \quad (3)$$

where $\mathbf{u} = (u, v)$ denotes the horizontal wind velocity vector. This feature is appended as an additional channel to the baseline input tensor, yielding $\mathbf{X} \in \mathbb{R}^{5 \times H \times W \times D}$.

Polar Wind Encoding. We also investigate replacing the Cartesian velocity components (u, v) with a polar encoding. The two velocity channels are transformed into three channels representing wind speed and direction:

$$s = \sqrt{u^2 + v^2}, \quad \cos \theta = \frac{u}{s}, \quad \sin \theta = \frac{v}{s}. \quad (4)$$

This formulation decouples wind speed from direction, which may provide a more natural factorization for learning terrain induced flow patterns. This encoding adds one channel to the baseline, yielding $\mathbf{X} \in \mathbb{R}^{5 \times H \times W \times D}$. When the polar encoding is combined with the flow incidence angle α (Section 3.1), the same discontinuity consideration applies: the scalar α has values near 0° and 360° that are numerically distant despite being geometrically equivalent.

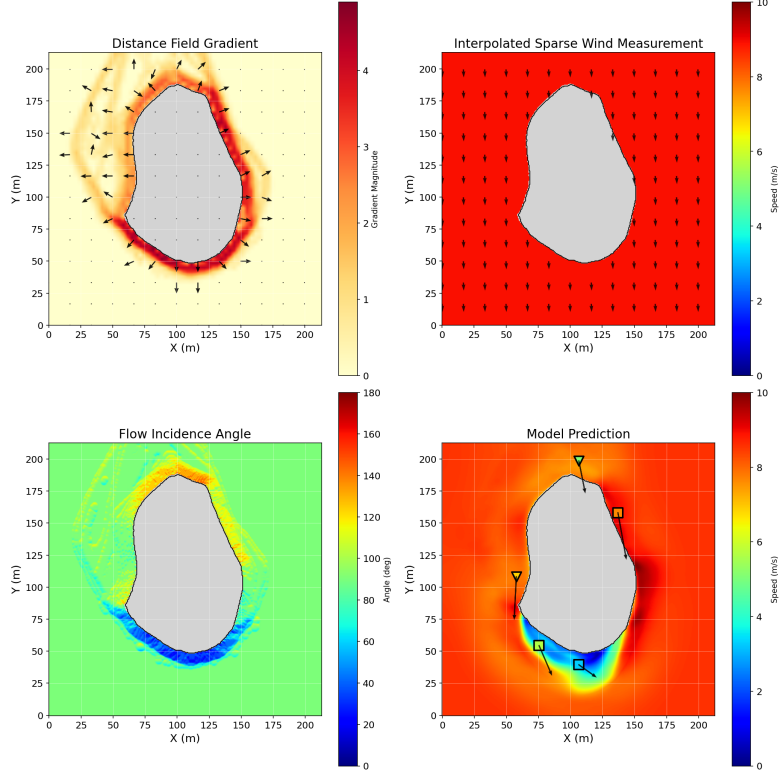


Figure 2: Horizontal slice through Bolund Hill. (A) Distance field gradient with arrows indicating the direction of steepest ascent from the terrain surface. (B) Interpolated sparse wind measurements from subset of Bolund hill mast observations. (C) Flow incidence angle (α feature), with low values (blue) indicating leeward regions and high values (yellow/red) marking windward regions. (D) Predicted wind speed from the α -augmented model. Markers denote measurement masts (squares: at slice height; triangles: above/below), with arrows indicating measured wind direction. The α feature captures flow characteristics including leeward deceleration, corner acceleration, and windward stagnation.

Following prior work on continuous angular representations for neural networks [35], we replace α with its sine/cosine projection $(\cos \alpha, \sin \alpha)$ to avoid this discontinuity, resulting in $\mathbf{X} \in \mathbb{R}^{7 \times H \times W \times D}$.

4. Experiments

4.1. Training and Test Data

We use the CFD wind dataset introduced in WindSeer [2]. The dataset consists of steady-state Reynolds-averaged Navier–Stokes (RANS) simulations with a $k\text{--}\epsilon$ turbulence closure, computed using OpenFOAM over 563 terrain patches extracted from Swiss digital elevation maps. Each $1.5 \text{ km} \times 1.5 \text{ km}$ patch was simulated at multiple wind speeds and directions, yielding 7,285 converged flow solutions. The solutions are resampled onto a regular Cartesian grid with 16.5 m horizontal and 11.5 m vertical resolution.

For training, we extract 64^3 subdomains with random translations and rotations as a geometric data augmentation. To emulate UAV flight paths, sparse wind measurements are sampled along piecewise-linear trajectories through each subdomain.

We do not train or evaluate on the synthetic LiDAR-based dataset of Folk et al. [3], as neither the dataset nor the trained models have been publicly released.

4.2. Real-World Evaluation Data

Measurement campaigns. We follow the mast-evaluation setup of WindSeer [2] on three publicly available campaigns that together span an order of magnitude in terrain scale: Bolund Hill [36] (Denmark), an 11 m coastal escarpment instrumented with 8 masts; Askervein Hill [37] (Scotland), a 116 m gently sloped hill with 14 masts; and Perdigão [38] (Portugal), two parallel forested ridges approximately 300 m high instrumented with 38 masts. Each mast is equipped with sonic or cup anemometers at heights ranging from 2 to 100 m above ground, and the long averaging periods reduce short-term effects such as gusts, yielding cleaner input measurements. We evaluate on this data; per-campaign inference and comparison counts are summarised in Table 2.

UAV flight campaigns. We follow the same UAV-evaluation setup as WindSeer [2]. Wind measurements were collected by multiple fixed-wing UAVs flown simultaneously over mountainous terrain in Switzerland: Chasseral in the Jura Mountains (3 UAVs), and Oberalppass and Gotthardpass in the Swiss Alps (2 UAVs each). The networks are evaluated along the recorded flight trajectories, with the on-board wind estimate (sampled at 50 Hz) smoothed by a ~ 20 s moving-average filter prior to use.

4.3. Model Architectures

We evaluate the proposed input features on four neural network architectures (3DNN, U-Net, FNO, U-FNO) spanning both convolutional and spectral paradigms. Additionally, we include larger architectures to assess whether the features improve the performance of both smaller and larger models in a similar manner.

3DNN. We adopt the 3DNN architecture introduced as WindSeer [2], a three-dimensional encoder-decoder based on the U-Net design [39]. The encoder compresses the input into a latent representation, processed in the bottleneck using fully connected layers before being passed to the decoder. Spatial resolution is restored via linear interpolation followed by convolutional layers, rather than transposed convolutions.

Fully Convolutional U-Net. This variant replaces the linear bottleneck with a fully convolutional bottleneck, preserving the spatial structure of the latent representation and enabling the network to model spatial correlations at the coarsest resolution. The decoder restores resolution using transposed convolutions.

Fourier Neural Operator (FNO). The FNO [6] learns mappings between function spaces by operating in the spectral domain. The input is first lifted to a higher-dimensional representation via a fully connected layer, then passed through multiple Fourier layers. Each Fourier layer applies a learned linear transformation in frequency space, adds a local linear bypass, and applies a nonlinear activation. A final fully connected layer projects the output to the target dimension.

U-Fourier Neural Operator (U-FNO). The U-FNO [7] extends the FNO by introducing a U-Net-style convolutional path within the operator layers.

These hybrid layers combine global spectral operations with local convolutions to better preserve high-frequency spatial information, capturing both global and local flow features.

For each architecture, we train both baseline and feature-augmented variants under identical training configurations, differing only in the input representation. We denote variants as B (baseline), A (flow incidence angle), P (polar encoding), and A+P (combined). In total, we evaluate 16 models: four variants for each of the 3DNN, U-Net, FNO, and U-FNO architectures. This setup enables a systematic assessment of feature performance across different network architectures and model capacities.

4.4. Training

All models were trained for 3000 epochs using the Adam optimizer. For the 3DNN architecture we attempted to replicate the training setup described in [2], namely a scaled ℓ_2 loss with a step learning-rate scheduler. All non-3DNN models employed a standard ℓ_2 loss with exponential learning-rate decay and architecture-specific normalization and learning-rate settings. Within each architecture, the training setup was held identical across the four feature variants (B, A, P, A+P), so the only thing that differs between a baseline and its feature-augmented counterpart is the input representation. Because the original WindSeer model weights had not been released alongside the public dataset at the time of this submission, the 3DNN baseline (B) reported in this paper is our retrained model rather than the published checkpoint.

4.5. Metrics

4.5.1. CFD synthetic data

For CFD-simulated flows we evaluate predictions of the time-averaged wind speed $\bar{U}$ and turbulent kinetic energy $\bar{k}$ over the full 3D prediction domain. Let $s \in \{1, \dots, S\}$ index samples and $i \in \{1, \dots, N\}$ spatial locations within a sample. We report a per-sample relative mean absolute error and an absolute maximum error,

$$y_{\text{MAE,rel}} = \frac{1}{S} \sum_{s=1}^S \frac{\frac{1}{N} \sum_{i=1}^N |\hat{y}_{s,i} - y_{s,i}|}{\frac{1}{N} \sum_{i=1}^N |y_{s,i}|} \times 100\%, \quad (5)$$

$$y_{\text{MME}} = \frac{1}{S} \sum_{s=1}^S \max_i |\hat{y}_{s,i} - y_{s,i}|, \quad (6)$$

where $y \in \{U, K\}$. MAE_{rel} measures the average per-sample error over the domain (in %), while MME captures the worst-case point-wise deviation per sample (in input units). Table 1 reports the four resulting columns in the order U_{MME} , $U_{\text{MAE,rel}}$, K_{MME} , $K_{\text{MAE,rel}}$ (wind-speed metrics first, then turbulent-kinetic-energy metrics).

4.5.2. UAV campaigns

For UAV flight experiments, metrics are computed along the recorded trajectories. We evaluate the horizontal wind speed W_h and the vertical wind speed W_z , reporting for each component the mean absolute error (MAE, in m/s) and the Pearson correlation coefficient. These four metrics correspond directly to the four data columns of Tables 3 and 4 (MAE for W_h , MAE for W_z , correlation for W_h , correlation for W_z).

4.5.3. Measurement campaigns

For ground-based weather-mast measurements we evaluate the horizontal wind speed W_h and the vertical wind speed W_z , reporting for each component the mean absolute error (MAE, in m/s) and the Pearson correlation coefficient. These match the four data columns of Table 5 (MAE Hor, Corr Hor, MAE Wz, Corr Wz).

4.6. Predicting CFD-Simulated Flows

We evaluated all variants on held-out CFD simulations over unseen terrains. Since the original test dataset does not include sparse trajectory wind measurement locations, we randomly sampled UAV measurement trajectories for evaluation. To better assess model performance we additionally extracted CFD simulations over more complex terrains, as a large portion of the original dataset was computed over predominantly flat surfaces. In total, we evaluated 50 randomly sampled UAV trajectories per CFD case, resulting in 39,700 trajectory-based test samples across 794 CFD simulations. All models were evaluated on the same set of trajectories, and the resulting performance metrics are summarized in Table 1.

Across the four architectures the proposed features consistently improve turbulence prediction while leaving wind-speed metrics close to baseline. The clearest single case is the compact 3DNN, where the flow incidence angle improves every metric column over baseline. On the larger architectures the flow incidence angle variant continues to match or improve the baseline on most metrics, while the polar encoding and the combined variant trade a

Table 1: CFD test set evaluation. The four metric columns are, in order: U_{MME} , the absolute maximum point error of the wind speed (worst-case error per sample, averaged across samples, in input units); $U_{\text{MAE,rel}}$, the per-sample relative mean absolute error of the wind speed (in %); and the same two quantities for the turbulent kinetic energy, K_{MME} and $K_{\text{MAE,rel}}$. Lower is better for all four. Within each architecture block the gray row is the baseline (B, four input channels); the three rows below it are the flow incidence angle feature (A), the polar wind-velocity encoding (P), and the combination (A+P, seven input channels). Each non-baseline cell reports the absolute value with the $\Delta\%$ improvement over the same-architecture baseline in parentheses (a bolded + marks an improvement); the best variant per column within each architecture is shown in bold. Yellow rows aggregate the per-architecture deltas as mean $\pm$ std across the four architectures.

Model	U_{MME}	$U_{\text{MAE,rel}}$ [%]	K_{MME}	$K_{\text{MAE,rel}}$ [%]
3DNN (5.5M parameters)				
3DNN_B	11.189	13.9	10.216	31.6
3DNN_A	10.862 (+2.9%)	12.3 (+11.3%)	10.127 (+0.9%)	28.1 (+11.3%)
3DNN_P	12.001 (-7.3%)	13.6 (+2.0%)	10.221 (-0.0%)	29.0 (+8.4%)
3DNN_A+P	11.235 (-0.4%)	12.9 (+7.0%)	10.233 (-0.2%)	28.7 (+9.4%)
FNO (402.6M parameters)				
FNO_B	10.890	16.6	12.152	40.0
FNO_A	10.935 (-0.4%)	16.7 (-0.7%)	11.478 (+5.5%)	38.8 (+3.0%)
FNO_P	11.413 (-4.8%)	17.0 (-2.8%)	10.915 (+10.2%)	39.3 (+1.6%)
FNO_A+P	11.646 (-6.9%)	17.4 (-4.9%)	10.728 (+11.7%)	38.3 (+4.1%)
UFNO (120.3M parameters)				
UFNO_B	10.397	12.9	10.093	29.2
UFNO_A	10.138 (+2.5%)	13.7 (-6.3%)	10.107 (-0.1%)	29.7 (-1.5%)
UFNO_P	10.007 (+3.8%)	14.4 (-11.5%)	9.847 (+2.4%)	30.1 (-2.9%)
UFNO_A+P	9.932 (+4.5%)	13.6 (-5.1%)	9.915 (+1.8%)	29.4 (-0.4%)
UNET (131.7M parameters)				
UNET_B	9.791	9.0	9.472	23.3
UNET_A	10.309 (-5.3%)	9.3 (-3.8%)	9.755 (-3.0%)	23.2 (+0.5%)
UNET_P	9.595 (+2.0%)	9.6 (-6.5%)	9.083 (+4.1%)	21.9 (+5.8%)
UNET_A+P	9.538 (+2.6%)	9.4 (-4.6%)	9.039 (+4.6%)	21.6 (+7.1%)
avg Δ A vs B	-0.1 $\pm$ 3.3%	+0.1 $\pm$ 6.8%	+0.8 $\pm$ 3.1%	+3.3 $\pm$ 4.8%
avg Δ P vs B	-1.6 $\pm$ 4.6%	-4.7 $\pm$ 5.0%	+4.2 $\pm$ 3.8%	+3.2 $\pm$ 4.3%
avg Δ A+P vs B	-0.1 $\pm$ 4.3%	-1.9 $\pm$ 5.1%	+4.5 $\pm$ 4.5%	+5.1 $\pm$ 3.6%

small wind-speed regression for the largest TKE gains; the U-Net combined variant shows the cleanest improvement among the larger networks. The yellow rows confirm the pattern in aggregate and identify the flow incidence angle as the safest single choice: it improves both TKE metrics on average while leaving wind-speed metrics essentially unchanged, whereas the polar encoding and the combined variant deliver larger TKE gains at a measurable wind-speed cost.

4.7. Real-world evaluations

We evaluate the four architectures on two complementary real-world data sources: time-averaged in-flight UAV wind measurements collected over Swiss alpine terrain, and near-surface anemometer-mast time series from three published wind-engineering campaigns. The two sources span different regimes of the wind field, of the terrain scale, and of the underlying flow physics; they therefore exercise the proposed features under different assumptions about what makes a learned wind-field predictor generalise. Table 2 summarises the evaluation density and mean-wind regime of each data source.

Table 2: Real-world evaluation density. For UAV flights, one inference is run per source loiter; validation points are the simultaneously sampled validation loiters from the remaining UAVs in the same flight. For mast campaigns, one inference is run per mast per case using that mast as the source measurement; validation points are the wind measurements from the remaining masts in the same case. The combined column reports the average number of validation points per inference and, in parentheses, the cumulative count over all inferences per network. Mean wind speed $\bar{U}$ is averaged over the evaluation domain (top-10 surface-mast values for mast campaigns, to suppress lee-side sheltered masts).

Source / group	Campaign extent	Inferences per network	Validation points per inference (total)	$\bar{U}$ [m/s]
UAV Chasseral	3 flights, 3 UAVs each	40	13.7 (546)	3.0
UAV Oberalppass+Gotthardpass	2 flights, 2 UAVs each	11	6.6 (73)	6.1
Perdigão masts (ISFS)	19 cases, ~37 masts	722	129 (93,138)	7.2
Askervein masts	13 cases, ~15 masts	211	27 (5,632)	14.0
Bolund masts	4 cases, ~9 masts	39	16 (636)	10.8

4.7.1. UAV flight campaigns

Tables 3 and 4 report flight-trajectory evaluation, with the cases split by mean wind speed: Chasseral at ~3 m/s and Oberalppass+Gotthardpass at ~6 m/s.

Table 3: UAV flight evaluation on Chasseral flights. The four metric columns are the mean absolute error (MAE, in m/s) and the Pearson correlation coefficient for the horizontal wind speed W_h and the vertical wind speed W_z ; lower MAE is better, higher correlation is better. Within each architecture block the gray row is the baseline (B, four input channels); the three rows below are the flow incidence angle (A), the polar wind-velocity encoding (P), and the combination (A+P, seven input channels). Each non-baseline cell reports the absolute value with the Δ over the same-architecture baseline in parentheses, expressed as a percentage for the MAE columns and as an absolute change for the correlation columns (a bolded + marks an improvement); the best variant per column within each architecture is shown in bold. Yellow rows aggregate the per-architecture deltas as mean $\pm$ std across the four architectures, in absolute units.

Model	MAE [m/s]		Correlation	
	W_h	W_z	W_h	W_z
3DNN (5.5M parameters)				
3DNN_B	0.56	0.26	0.48	0.86
3DNN_A	0.51 (+8.0%)	0.39 (-47.9%)	0.54 (+0.05)	0.85 (-0.01)
3DNN_P	1.27 (-127.2%)	0.48 (-82.4%)	0.49 (+0.01)	0.87 (+0.01)
3DNN_A+P	1.33 (-138.5%)	0.38 (-43.9%)	0.49 (+0.00)	0.90 (+0.04)
FNO (402.6M parameters)				
FNO_B	0.55	0.38	0.59	0.67
FNO_A	0.56 (-0.6%)	0.34 (+10.7%)	0.30 (-0.29)	0.73 (+0.06)
FNO_P	0.59 (-6.3%)	0.45 (-18.1%)	0.52 (-0.07)	0.05 (-0.62)
FNO_A+P	2.22 (-301.7%)	0.36 (+4.5%)	0.40 (-0.19)	0.78 (+0.11)
UFNO (120.3M parameters)				
UFNO_B	0.58	0.38	0.40	0.68
UFNO_A	0.55 (+5.5%)	0.38 (+1.9%)	0.22 (-0.18)	0.68 (-0.00)
UFNO_P	0.55 (+5.9%)	0.40 (-4.4%)	0.46 (+0.06)	0.75 (+0.07)
UFNO_A+P	2.09 (-258.7%)	0.30 (+22.2%)	-0.20 (-0.60)	0.77 (+0.09)
UNET (131.7M parameters)				
UNET_B	0.53	0.30	0.28	0.82
UNET_A	0.54 (-2.5%)	0.35 (-14.3%)	0.49 (+0.21)	0.88 (+0.06)
UNET_P	0.53 (+0.8%)	0.37 (-22.0%)	0.28 (+0.00)	0.78 (-0.04)
UNET_A+P	2.12 (-300.2%)	0.35 (-15.5%)	0.44 (+0.16)	0.86 (+0.04)
avg Δ A vs B	+0.02 $\pm$ 0.02	-0.03 $\pm$ 0.06	-0.05 $\pm$ 0.20	+0.03 $\pm$ 0.03
avg Δ P vs B	-0.18 $\pm$ 0.31	-0.09 $\pm$ 0.07	-0.00 $\pm$ 0.04	-0.15 $\pm$ 0.28
avg Δ A+P vs B	-1.39 $\pm$ 0.36	-0.01 $\pm$ 0.07	-0.16 $\pm$ 0.28	+0.07 $\pm$ 0.03

Table 4: UAV flight evaluation on Oberalppass+Gotthardpass flights. Same conventions as Table 3.

Model	MAE [m/s]		Correlation	
	W_h	W_z	W_h	W_z
3DNN (5.5M parameters)				
3DNN_B	0.95	0.58	-0.67	0.66
3DNN_A	1.21 (-27.3%)	0.43 (+25.6%)	0.12 (+0.79)	0.85 (+0.19)
3DNN_P	1.38 (-45.8%)	0.63 (-8.5%)	0.27 (+0.94)	0.24 (-0.42)
3DNN_A+P	1.48 (-55.8%)	0.57 (+1.9%)	0.24 (+0.91)	0.19 (-0.47)
FNO (402.6M parameters)				
FNO_B	0.89	0.37	0.18	0.68
FNO_A	0.92 (-2.7%)	0.39 (-6.4%)	-0.22 (-0.40)	0.31 (-0.37)
FNO_P	0.84 (+6.5%)	0.53 (-44.9%)	0.06 (-0.12)	0.04 (-0.64)
FNO_A+P	5.47 (-511.6%)	0.43 (-15.9%)	-0.12 (-0.30)	0.05 (-0.63)
UFNO (120.3M parameters)				
UFNO_B	1.04	0.46	-0.12	0.29
UFNO_A	0.90 (+13.6%)	0.41 (+9.8%)	0.07 (+0.19)	0.07 (-0.22)
UFNO_P	0.84 (+19.0%)	0.47 (-2.2%)	0.14 (+0.26)	0.36 (+0.07)
UFNO_A+P	5.59 (-436.3%)	0.41 (+9.2%)	-0.16 (-0.04)	0.45 (+0.16)
UNET (131.7M parameters)				
UNET_B	1.04	0.48	-0.17	0.56
UNET_A	0.95 (+9.0%)	0.86 (-80.8%)	-0.31 (-0.14)	0.43 (-0.13)
UNET_P	0.88 (+16.0%)	0.59 (-23.1%)	0.24 (+0.41)	0.20 (-0.36)
UNET_A+P	5.21 (-398.8%)	0.43 (+9.8%)	0.38 (+0.56)	0.29 (-0.27)
avg Δ A vs B	-0.01 $\pm$ 0.20	-0.05 $\pm$ 0.29	+0.11 $\pm$ 0.80	-0.13 $\pm$ 0.60
avg Δ P vs B	-0.00 $\pm$ 0.36	-0.08 $\pm$ 0.14	+0.37 $\pm$ 0.61	-0.34 $\pm$ 0.57
avg Δ A+P vs B	-3.45 $\pm$ 1.72	+0.01 $\pm$ 0.08	+0.28 $\pm$ 0.78	-0.30 $\pm$ 0.47

In the low-wind Chasseral regime, absolute errors are less informative for assessing model performance: horizontal MAE values cluster around 0.55 m/s across architectures and features, and per-feature differences mostly remain within the architecture-to-architecture spread.

In the higher-wind Oberalppass+Gotthardpass regime the picture is cleaner in horizontal metrics. The flow incidence angle holds horizontal MAE at the baseline level and improves horizontal-wind correlation on average; this is the most consistent benefit the features deliver in flight evaluation, though the spread across architectures is large enough that we present the result as a tendency rather than a uniform improvement. The polar encoding shows a larger mean correlation gain but with high variance across architectures, and the combined variant produces large horizontal MAE outliers above 5 m/s on FNO, U-FNO, and U-Net. The polar encoding and the combined variant therefore exhibit clear failure modes and are not as robust as the flow incidence angle alone.

4.7.2. Near-surface mast measurements

We evaluate on near-surface mast measurements from three campaigns: Perdigão (Portugal, ISFS sensor network), Askervein Hill (Scotland) and Bolund Hill (Denmark). The masts sample fixed near-surface heights, so the evaluation reflects a portion of the predicted field biased toward the ground rather than the trajectory-averaged regime of §4.7.1. The three campaigns also differ markedly in terrain scale relative to the training data: Perdigão at native ~ 8 m grid resolution sits closest to the 16.5 m horizontal / 11.5 m vertical training resolution (over 1.5 km patches), while Askervein (~ 4 m native) and Bolund (~ 0.55 m native) are progressively smaller hills outside the spatial scale at which the networks were trained. Perdigão also dominates by evaluation weight: $\sim 93\%$ of total comparisons across the three campaigns (93,138 of $\sim 99,406$) come from Perdigão (Table 2).

Table 5: Architecture-averaged mast-measurement results. The four metric columns are the mean absolute error (MAE, in m/s) and the Pearson correlation coefficient for the horizontal wind speed W_h and the vertical wind speed W_z ; lower MAE is better, higher correlation is better. Each row reports the feature-vs-baseline delta averaged across the four architectures (3DNN, FNO, U-FNO, U-Net) as mean $\pm$ std, where positive values indicate improvement over the same-architecture baseline (B). The three variants compared against the baseline are the flow incidence angle (A), the polar wind-velocity encoding (P), and the combination (A+P). Cells in bold mark a positive on-average improvement. The table is split into three blocks by campaign group: Perdigão at the top, Bolund+Askervein in the middle, and a per-inference weighted average of the two groups at the bottom (weights 722 and 250, proportional to the number of inferences per network reported in Table 2). Per-architecture values are reported in Appendix A.

Variant	MAE Hor	Corr Hor	MAE Wz	Corr Wz
Perdigão				
avg Δ A vs B	+0.18 $\pm$ 0.26	+0.03 $\pm$ 0.04	+0.09 $\pm$ 0.22	-0.10 ± 0.20
avg Δ P vs B	+0.40 $\pm$ 0.52	+0.11 $\pm$ 0.27	+0.14 $\pm$ 0.26	-0.13 ± 0.17
avg Δ A+P vs B	+0.31 $\pm$ 0.63	+0.10 $\pm$ 0.30	+0.11 $\pm$ 0.27	-0.13 ± 0.12
Bolund + Askervein				
avg Δ A vs B	-0.10 ± 0.60	+0.00 $\pm$ 0.10	-0.03 ± 0.06	-0.07 ± 0.13
avg Δ P vs B	-4.29 ± 1.79	+0.09 $\pm$ 0.22	+0.06 $\pm$ 0.08	-0.15 ± 0.13
avg Δ A+P vs B	-4.23 ± 1.92	-0.05 ± 0.36	+0.03 $\pm$ 0.08	+0.09 $\pm$ 0.14
Per-inference weighted average across both campaign groups				
avg Δ A vs B	+0.11 $\pm$ 0.36	+0.02 $\pm$ 0.05	+0.06 $\pm$ 0.18	-0.10 ± 0.16
avg Δ P vs B	-0.81 ± 0.33	+0.11 $\pm$ 0.29	+0.12 $\pm$ 0.24	-0.14 ± 0.18
avg Δ A+P vs B	-0.85 ± 0.32	+0.06 $\pm$ 0.35	+0.09 $\pm$ 0.25	-0.07 ± 0.08

Table 5 reports the architecture-averaged feature-vs-baseline deltas; the per-architecture tables are in Appendix A. On Perdigão all three feature variants improve horizontal-wind MAE, horizontal-wind correlation, and vertical-wind MAE on average; the polar encoding and the combined variant give the

largest horizontal improvements, while the flow incidence angle delivers a smaller but more uniform gain. The standout single result is the U-Net combined variant, which roughly halves the vertical-wind MAE (from 1.06 to 0.50 m/s) and lifts horizontal correlation from 0.13 to 0.69.

On the Bolund+Askervein group the polar encoding collapses on horizontal MAE on average and the combined variant inherits the same failure, with values jumping to roughly 8.5 m/s on FNO, U-FNO and U-Net. The flow incidence angle, by contrast, holds within noise on average. On the per-inference weighted average across both campaign groups (Perdigão $\times 0.74$, Bolund+Askervein $\times 0.26$) the polar encoding still delivers a horizontal-correlation gain but its horizontal MAE remains negative on average due to the small-domain collapse, while the flow incidence angle keeps both horizontal metrics near baseline and provides modest vertical-wind MAE gains.

4.7.3. Discussion

Real-world results across the UAV and mast campaigns reveal a consistent ordering between the three feature variants. Two observations help interpret it: the polar encoding’s sensitivity to terrain-scale shift, visible most clearly on the small-domain mast campaigns; and the spatial structure of where the gains arise, visible on the per-mast Perdigão breakdown.

Feature ordering and the scale-shift mechanism.. Across the three evaluation settings the flow incidence angle is the most robust of the three feature variants, modestly improving or matching baseline on essentially every metric. The polar encoding delivers the largest correlation gains where it applies but is sensitive to terrain-scale shift, collapsing on the small-domain Bolund and Askervein cases, and the combined variant inherits both behaviours. Figure 3 visualises the scale mismatch underlying this collapse: a single 16.6 m training cell contains 30×30 Bolund inference cells, 4×4 Askervein cells, and 2×2 Perdigão cells. We interpret the failure of the polar encoding on the small-scale campaigns as a scale-shift effect: the polar form $(s, \cos \theta, \sin \theta)$ embeds wind magnitude and direction in the training-domain Cartesian frame, so the network learns a decomposition tied to that scale and does not transfer to substantially smaller domains. The flow incidence angle, by contrast, is a dimensionless quantity invariant under rescaling of both terrain and flow and is therefore decoupled from the training scale; we offer this as a working interpretation since the present experiments do not isolate the mechanism directly.

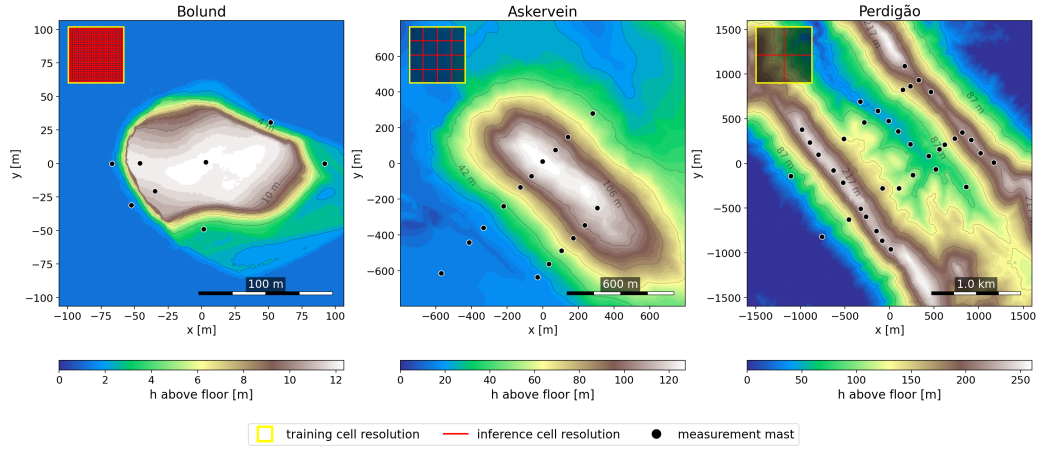


Figure 3: Native terrain extent of the three measurement campaigns at full resolution, each with its own elevation colormap and Google-Maps style scale bar in the bottom-right corner. The top-left inset on each panel shows the cell-size mismatch between the campaign-native inference grid (red sub-grid) and the network’s training cell (yellow box, 16.6 m): a single training cell contains 30×30 Bolund inference cells, 4×4 Askervein cells, and 2×2 Perdigão cells. Black dots mark the measurement masts. Bolund and Askervein sit one to two orders of magnitude below the spatial scale at which the networks were trained, while Perdigão is closest to the training distribution.

Per-mast spatial structure on Perdigão.. Figures 4 and 5 resolve the per-mast B–A delta on Perdigão. At each mast position the plotted value is the change in 3D-vector MAE evaluated against the remaining masts in each case when that mast’s measurements are used as the trajectory input, so a positive (red) delta marks a mast at which the network produces a lower error under the flow incidence angle than under the baseline. The yellow arrow marks the mean wind direction across the 19 cases. The lower-error region concentrates on the lee side of the two parallel ridges, while the gain is smaller on the ridge tops where the flow closely follows the surface gradient. The pattern agrees across the four architectures. This slightly improves on a reported weakness of the WindSeer baseline [2], where lee-side input measurements degraded the prediction at the other masts; the flow incidence angle supplies an explicit windward-versus-leeward inductive bias, so the network does not have to recover that distinction from the sparse inputs alone.

Vertical-wind prediction.. Across all evaluations the proposed features improve horizontal-wind metrics consistently but leave vertical-wind metrics closer to baseline; in particular, vertical-wind correlation slightly degrades on the mast campaigns. The flow incidence angle as defined in this work is computed against the horizontal terrain gradient and is therefore most informative for in-plane flow deflection. A natural extension is to define an analogous incidence angle against the vertical component of the terrain gradient, which would target the windward updraft and lee-side downdraft signal directly; we expect this variant to recover some of the vertical-wind metric headroom that the present features leave on the table.

4.8. Computational cost

Table 6 reports inference times on an embedded NVIDIA Jetson Spark platform, representative of onboard UAV hardware. Inference was evaluated on grid resolutions of $384 \times 384 \times 192$ for the 3DNN and $192 \times 192 \times 96$ for U-Net, FNO, and U-FNO, corresponding to the same physical domain size at different spatial resolutions. Across all four architectures, increasing the input from 4 (baseline) to 7 (A+P) channels leaves inference time effectively unchanged, confirming that the proposed features add negligible computational overhead. The compact 3DNN at ~ 30 ms per forward pass is fast enough for onboard control-rate inference, and the larger architectures remain within the budgets used by recent wind-aware UAV planners [1]. Figure 6 shows that the proposed features also maintain or slightly improve

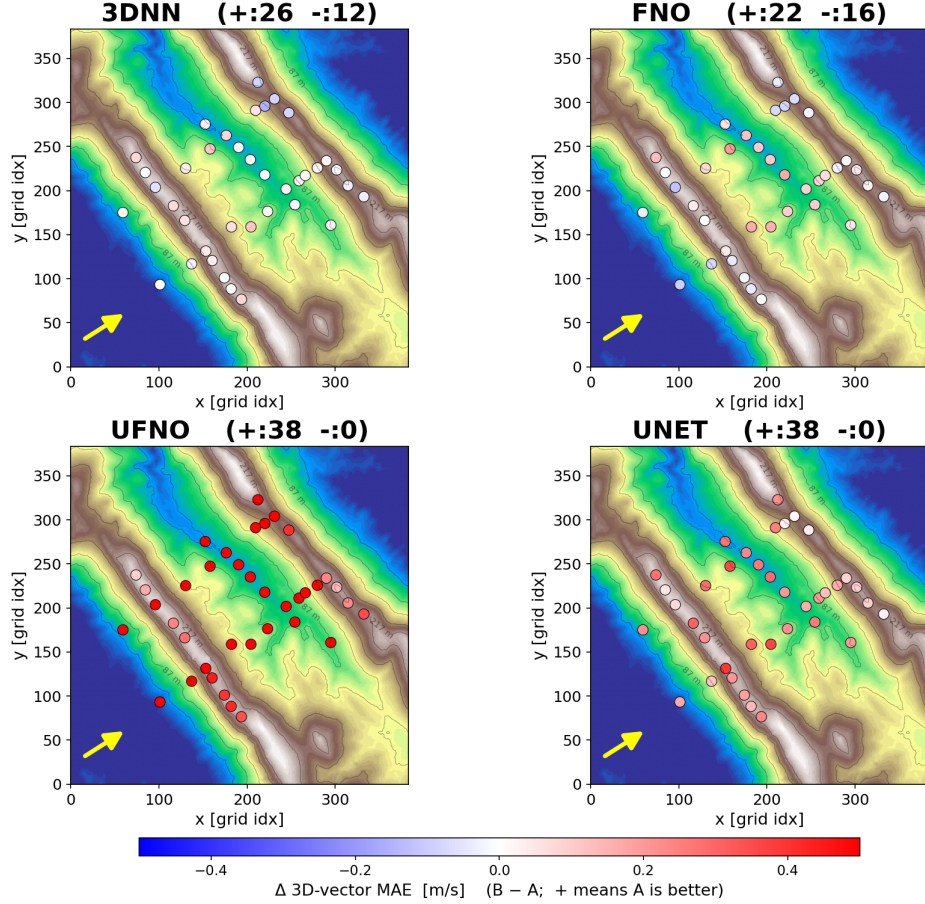


Figure 4: Per-mast spatial structure of the flow-incidence-angle gain on Perdigão, averaged across all 19 ISFS cases. **B** denotes the same-architecture baseline (four-channel input); **A** denotes the variant with the flow incidence angle as an additional input (five channels). At each mast position the plotted value is the B–A change in 3D-vector MAE evaluated against the remaining masts in each case when that mast’s measurements are used as the trajectory input: a positive delta (red) marks a mast at which the network reaches a lower error under A, and a negative delta (blue) marks a higher error. Values are absolute deltas in m/s, clipped to ± 0.5 m/s. The number of mast positions with positive and negative delta is annotated in each panel title. The yellow arrow (identical on every panel) shows the dominant measurement-derived wind across the 19 cases (top-10 masts each, mean magnitude 2.1 m/s). Black contour lines mark terrain elevation.

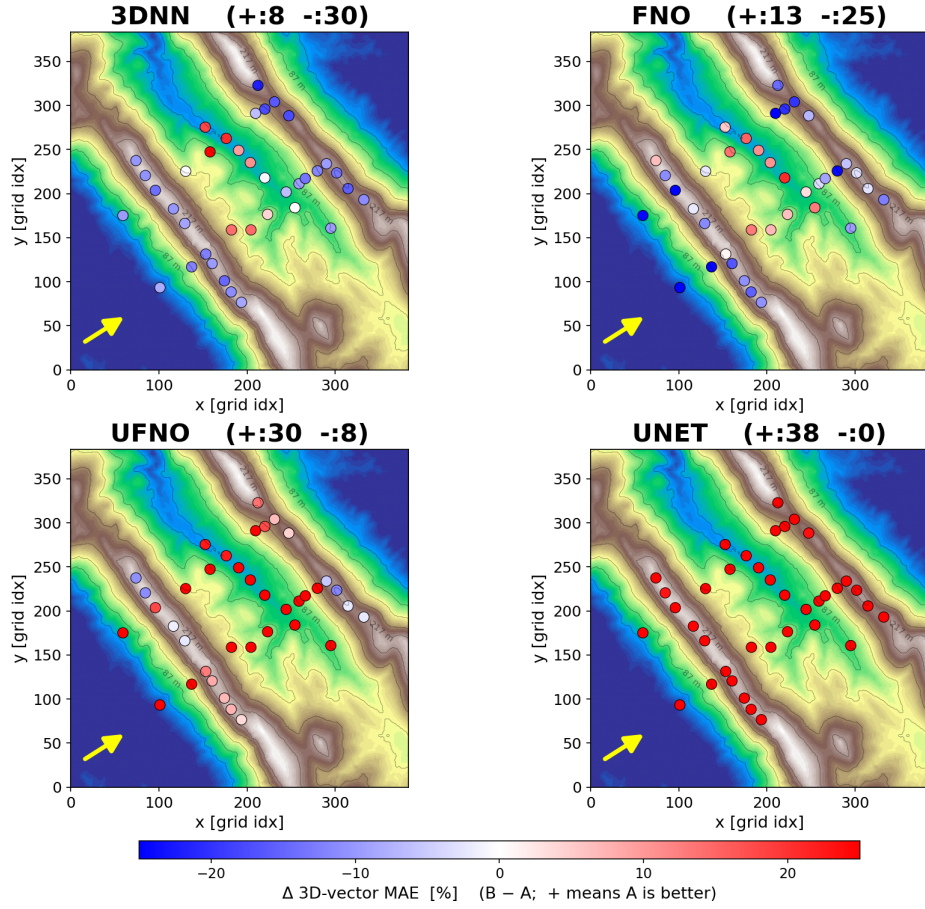


Figure 5: Per-mast spatial structure on Perdigão, relative version. Same conventions as Fig. 4, but reporting the relative 3D-vector MAE delta in percentage points, clipped to $\pm 25\%$.

Table 6: Inference time (ms) for neural wind field predictors on small (1 km^2) and large (36 km^2) domains. The B column reports the baseline model with 4 input channels, the A+P column reports the variant with 7 input channels. All measurements were performed on an NVIDIA Jetson Spark.

Model	Params	1 km ² Domain		36 km ² Domain	
		B	A+P	B	A+P
UNet	131.7M	339	339	4545	4531
FNO	402.6M	181	205	1846	1723
UFNO	120.3M	409	409	5825	5796
3DNN	5.5M	29	30	3147	3212

training convergence, with feature-augmented models reaching comparable loss values in fewer epochs, particularly during early training.

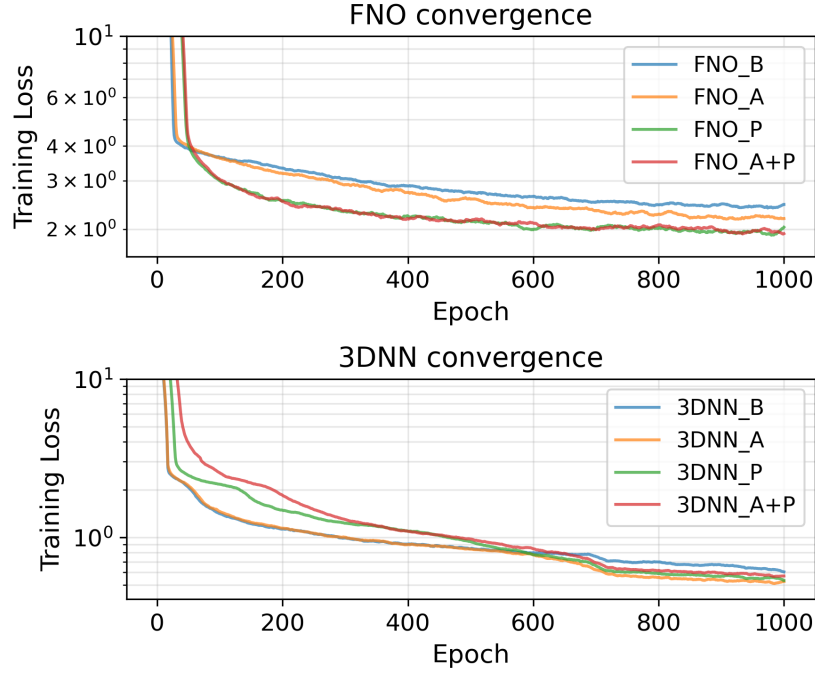


Figure 6: Training loss curves for FNO and 3DNN architectures across all feature variants. The proposed features accelerate early convergence, with feature-augmented models reaching lower loss values during the first third of training.

5. Conclusion

We presented two terrain-aware input features for learning-based wind field prediction: a flow incidence angle that encodes the angular relationship between the incoming wind and the local terrain gradient, and a polar wind-velocity encoding that replaces the Cartesian wind components with magnitude and angular components. We evaluated four feature variants (B baseline, A flow incidence angle, P polar encoding, A+P combined) across four architectures (3DNN, FNO, U-FNO, U-Net) on CFD-simulated flows, UAV flight trajectories, and three near-surface mast measurement campaigns.

Across this evaluation the flow incidence angle emerges as the safest single choice. On CFD it improves both turbulent-kinetic-energy metrics on average while leaving wind-speed metrics essentially unchanged, on UAV flight trajectories it holds horizontal MAE at the baseline level and gives the most consistent horizontal-correlation gain, and on the in-distribution Perdigão mast campaign it improves all four metric averages while remaining within noise on the small-domain Bolund and Askervein campaigns. A per-mast breakdown on Perdigão further shows that the flow incidence angle slightly improves on a reported weakness of the WindSeer baseline [2], in which prediction quality degraded when the input measurement came from a lee-side mast. The polar encoding delivers larger absolute gains on CFD and on Perdigão but is sensitive to terrain-scale shift: when applied at terrain scales an order of magnitude smaller than the network’s training cell, it collapses the horizontal-MAE error by several m/s. The combined A+P variant inherits this scale sensitivity. Training the polar encoding across a range of terrain scales is a natural next step that we expect to recover its CFD-level gains under scale shift. The proposed features are architecture-agnostic, add no measurable inference overhead, and can be integrated into any existing wind-field prediction framework by extending the input channel count.

A natural future direction is to extend the proposed features to urban wind-field surrogates [19, 8], which target dense building geometries and consume different terrain representations (building footprints rather than continuous topography) and are therefore evaluated on disjoint datasets from the alpine setting considered here.

Acknowledgements

We thank the WindSeer authors [2] for preparing and publicly releasing the CFD training dataset used in this work. The access to the computational infrastructure of the OP VVV funded project CZ.02.1.01/0.0/0.0/16_019/0000765 “Research Center for Informatics” is also gratefully acknowledged.

CRedit authorship contribution statement

Michal Kasarda: Conceptualization, Methodology, Software, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Robert Bakhtigariyev:** Conceptualization, Methodology, Software, Investigation. **Karel Zimmermann:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This work was co-funded by the Grant Agency of the CTU in Prague under Project SGS26/075/OHK3/1T/13, the European Union under the project Robotics and Advanced Industrial Production (Roboprox, reg. no. CZ.02.01.01/00/22_008/0004590), and the Czech Science Foundation under Project 24-12360S.

Data availability

The implementation of the proposed flow incidence angle and polar wind-velocity encodings, the training scripts, and the trained model checkpoints for all four architectures will be made publicly available upon acceptance. The CFD training dataset, the UAV flight data, and the three near-surface mast measurement datasets used in this work are publicly available from their original providers.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI assistance to refurbish the manuscript text and to prototype parts of the code-base. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Per-architecture mast-measurement results

This appendix reports the full per-architecture results underlying the architecture-averaged summary in Table 5. Conventions are identical to the summary table: MAE in m/s (lower is better); correlation $\in [-1, 1]$ (higher is better); each non-baseline cell shows the absolute value with the same-architecture Δ in parentheses ($\Delta\%$ for MAE columns, absolute Δ for correlation columns; bold + indicates improvement over the same-architecture baseline). Yellow rows aggregate the per-architecture deltas as mean $\pm$ std across the four architectures.

Table A.7: Per-architecture mast results, Perdigo (in-distribution terrain scale, $\sim 93\%$ of evaluation weight). Native uncropped domain.

Model	MAE Hor	Corr Hor	MAE Wz	Corr Wz
3DNN (5.5M parameters)				
3DNN_B	3.09	0.39	0.51	0.31
3DNN_A	3.07 (+ 0.5%)	0.38 (-0.01)	0.52 (-2.3%)	0.27 (-0.05)
3DNN_P	3.41 (-10.4%)	0.44 (+ 0.05)	0.59 (-17.0%)	0.34 (+ 0.02)
3DNN_A+P	3.72 (-20.7%)	0.39 (+ 0.00)	0.66 (-30.0%)	0.38 (+ 0.06)
FNO (402.6M parameters)				
FNO_B	3.74	-0.07	0.48	0.32
FNO_A	3.70 (+ 1.2%)	-0.04 (+ 0.03)	0.56 (-16.3%)	-0.06 (-0.38)
FNO_P	3.03 (+ 19.2%)	0.10 (+ 0.18)	0.54 (-11.2%)	-0.09 (-0.41)
FNO_A+P	3.07 (+ 18.1%)	0.02 (+ 0.10)	0.54 (-11.3%)	0.05 (-0.27)
UFNO (120.3M parameters)				
UFNO_B	4.10	0.05	0.67	-0.01
UFNO_A	3.47 (+ 15.4%)	0.13 (+ 0.09)	0.70 (-4.4%)	0.16 (+ 0.17)
UFNO_P	3.07 (+ 25.1%)	-0.21 (-0.26)	0.53 (+ 20.3%)	-0.13 (-0.11)
UFNO_A+P	3.07 (+ 25.1%)	-0.22 (-0.27)	0.56 (+ 16.3%)	-0.14 (-0.13)
UNET (131.7M parameters)				
UNET_B	3.16	0.13	1.06	0.47
UNET_A	3.14 (+ 0.6%)	0.13 (-0.01)	0.58 (+ 44.8%)	0.31 (-0.15)
UNET_P	2.99 (+ 5.3%)	0.62 (+ 0.49)	0.49 (+ 53.3%)	0.43 (-0.04)
UNET_A+P	2.97 (+ 6.0%)	0.69 (+ 0.56)	0.50 (+ 52.2%)	0.27 (-0.19)
avg Δ A vs B	+ 0.18 $\pm$ 0.26	+ 0.03 $\pm$ 0.04	+ 0.09 $\pm$ 0.22	-0.10 $\pm$ 0.20
avg Δ P vs B	+ 0.40 $\pm$ 0.52	+ 0.11 $\pm$ 0.27	+ 0.14 $\pm$ 0.26	-0.13 $\pm$ 0.17
avg Δ A+P vs B	+ 0.31 $\pm$ 0.63	+ 0.10 $\pm$ 0.30	+ 0.11 $\pm$ 0.27	-0.13 $\pm$ 0.12

Table A.8: Per-architecture mast results, Bolund + Askervein (out-of-distribution small-scale terrain). Native uncropped domains.

Model	MAE Hor	Corr Hor	MAE Wz	Corr Wz
3DNN (5.5M parameters)				
3DNN_B	2.77	0.37	0.48	0.78
3DNN_A	2.80 (-1.1%)	0.39 (+ 0.02)	0.47 (+ 1.7%)	0.68 (-0.10)
3DNN_P	3.96 (-43.1%)	0.45 (+ 0.07)	0.46 (+ 4.2%)	0.75 (-0.03)
3DNN_A+P	3.68 (-33.0%)	0.42 (+ 0.05)	0.51 (-7.5%)	0.74 (-0.04)
FNO (402.6M parameters)				
FNO_B	3.23	0.03	0.69	0.34
FNO_A	3.03 (+ 6.2%)	0.11 (+ 0.08)	0.65 (+ 5.9%)	0.44 (+ 0.10)
FNO_P	8.60 (-166.8%)	0.27 (+ 0.25)	0.75 (-7.7%)	-0.02 (-0.36)
FNO_A+P	8.53 (-164.5%)	0.14 (+ 0.11)	0.75 (-8.3%)	0.65 (+ 0.31)
UFNO (120.3M parameters)				
UFNO_B	3.39	0.21	0.82	0.24
UFNO_A	2.87 (+ 15.3%)	0.29 (+ 0.08)	0.89 (-8.1%)	0.20 (-0.04)
UFNO_P	8.58 (-153.6%)	-0.04 (-0.25)	0.71 (+ 13.8%)	0.18 (-0.06)
UFNO_A+P	8.50 (-151.2%)	-0.46 (-0.66)	0.72 (+ 12.9%)	0.23 (-0.01)
UNET (131.7M parameters)				
UNET_B	2.99	0.27	0.81	0.60
UNET_A	4.06 (-36.0%)	0.10 (-0.17)	0.91 (-12.2%)	0.34 (-0.26)
UNET_P	8.37 (-180.5%)	0.58 (+ 0.31)	0.66 (+ 18.1%)	0.46 (-0.14)
UNET_A+P	8.58 (-187.3%)	0.56 (+ 0.29)	0.70 (+ 13.8%)	0.71 (+ 0.11)
avg Δ A vs B	-0.10 $\pm$ 0.60	+ 0.00 $\pm$ 0.10	-0.03 $\pm$ 0.06	-0.07 $\pm$ 0.13
avg Δ P vs B	-4.29 $\pm$ 1.79	+ 0.09 $\pm$ 0.22	+ 0.06 $\pm$ 0.08	-0.15 $\pm$ 0.13
avg Δ A+P vs B	-4.23 $\pm$ 1.92	-0.05 $\pm$ 0.36	+ 0.03 $\pm$ 0.08	+ 0.09 $\pm$ 0.14

Table A.9: Per-architecture mast results, unweighted average of Perdigão and Bolund+Askervein.

Model	MAE Hor	Corr Hor	MAE Wz	Corr Wz
3DNN (5.5M parameters)				
3DNN_B	2.93	0.38	0.49	0.55
3DNN_A	2.93 (-0.3%)	0.38 (+ 0.00)	0.49 (-0.3%)	0.47 (-0.08)
3DNN_P	3.68 (-25.8%)	0.44 (+ 0.06)	0.52 (-6.7%)	0.54 (-0.00)
3DNN_A+P	3.70 (-26.5%)	0.40 (+ 0.02)	0.58 (-19.1%)	0.56 (+ 0.01)
FNO (402.6M parameters)				
FNO_B	3.48	-0.02	0.59	0.33
FNO_A	3.36 (+ 3.5%)	0.04 (+ 0.06)	0.61 (-3.2%)	0.19 (-0.14)
FNO_P	5.82 (-66.9%)	0.19 (+ 0.21)	0.64 (-9.1%)	-0.05 (-0.38)
FNO_A+P	5.80 (-66.4%)	0.08 (+ 0.10)	0.64 (-9.5%)	0.35 (+ 0.02)
UFNO (120.3M parameters)				
UFNO_B	3.74	0.13	0.75	0.11
UFNO_A	3.17 (+ 15.3%)	0.21 (+ 0.08)	0.79 (-6.4%)	0.18 (+ 0.07)
UFNO_P	5.83 (-55.7%)	-0.13 (-0.26)	0.62 (+ 16.7%)	0.03 (-0.08)
UFNO_A+P	5.79 (-54.6%)	-0.34 (-0.47)	0.64 (+ 14.4%)	0.04 (-0.07)
UNET (131.7M parameters)				
UNET_B	3.07	0.20	0.93	0.53
UNET_A	3.60 (-17.2%)	0.11 (-0.09)	0.75 (+ 20.1%)	0.33 (-0.21)
UNET_P	5.68 (-84.9%)	0.60 (+ 0.40)	0.58 (+ 38.1%)	0.44 (-0.09)
UNET_A+P	5.77 (-87.9%)	0.62 (+ 0.42)	0.60 (+ 35.6%)	0.49 (-0.04)
avg Δ A vs B	+ 0.04 $\pm$ 0.39	+ 0.01 $\pm$ 0.07	+ 0.03 $\pm$ 0.09	-0.09 $\pm$ 0.10
avg Δ P vs B	-1.95 $\pm$ 0.71	+ 0.10 $\pm$ 0.24	+ 0.10 $\pm$ 0.16	-0.14 $\pm$ 0.14
avg Δ A+P vs B	-1.96 $\pm$ 0.72	+ 0.02 $\pm$ 0.32	+ 0.07 $\pm$ 0.17	-0.02 $\pm$ 0.04

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